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**NAAC
GRADE A+**
ACCREDITED UNIVERSITY

**Master of Computer Applications
Semester – III
Data Analytics for Business**

Final Project

Title: Amazon India Sales Dashboard – E-Commerce Business Intelligence Solution

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Amazon India Sales Dashboard

Developing the Amazon India Sales Dashboard

GitHub Link: <https://github.com/aathish592004-commits/data-analytics>

Power BI Service Link: https://app.powerbi.com/links/_yAQOc3p56?ctid=6bab7765-c387-4d0d-9be7-7f5679251594&pbi_source=linkShare

Problem Statement and Data Collection:

Given Problem Statement:

As a Business Intelligence Consultant for a growing organization, the task is to develop an end-to-end business analytics solution using a dataset of choice, in order to support data-driven business decisions.

Project-Specific Problem Statement:

The objective of this project is to develop a Business Intelligence solution for the e-commerce retail domain using Amazon India order-level sales data. The dashboard analyses total revenue, gross profit and profit margin, order volume and fulfilment performance, on-time shipping and product returns, payment success and refund behaviour, customer growth, and category/product-level performance across US states. The solution helps a retail operations, finance, or category-management team identify high-performing states and product categories, monitor order-to-delivery health, control payment and refund risk, and generate actionable insights to support inventory planning, logistics, and revenue growth.

Data Description:

Dataset used: an Amazon India order-level relational dataset (amazon_db) loaded into Power BI via Power Query, consisting of multiple related tables. Key tables and columns include:

- orders — order_id, customer_id, order_status, order_year, order_date
- orders_items — order_id, product_id, quantity, Total_revenue (line-level revenue)
- products — product_id, product_name, category_name, cogs (cost of goods sold)
- payments — payment_id, order_id, payment_status (Payment Succeeded / Payment Failed / Refunded)
- shipping — order_id, ship_date, return_date (blank when a product has not been returned)
- customers — customer_id, state, city, join details
- inventory — product_id, stock (current stock-on-hand quantity)

Business Problem:

Retail operations and finance teams need a fast way to see how the business is actually performing — not just total revenue, but profitability after cost of goods sold, how reliably orders are fulfilled and shipped on time, how much revenue is being lost to refunds and returns, and which states, categories and products are driving results. This project builds a single interactive dashboard that answers that question at both the company-wide level and the state/category/product level.

Data Cleaning & Transformation (Power Query):

The source tables were connected in Power BI via Get Data, and the following transformations were applied in Power Query Editor before loading to the model:

- Removed duplicate rows on order_id / order_id + product_id to avoid double-counting orders and line-item revenue.
- Changed data types: quantity, Total_revenue, cogs and stock to Whole Number / Decimal; order_date, ship_date and return_date to Date.
- Handled blank return_date values as "not returned" rather than errors, so return-rate measures could safely test for a non-blank date.
- Standardised payment_status text values (e.g. "Payment Succeeded", "Payment Failed", "Refunded") to remove stray casing and whitespace inconsistencies.
- Trimmed and cleaned text columns (state, category_name, product_name) using Trim/Clean steps to remove stray whitespace.
- Renamed columns to business-friendly names (e.g. Total_orders, Total_revenue, total_cogs) for cleaner DAX formulas and clearer visuals.
- Verified referential integrity between orders, orders_items, products, payments, shipping, customers and inventory, and confirmed no fully blank rows remained before closing & loading.

Data Modelling & Relationship:

The model uses a star-schema style design built around the cleaned Amazon order data:

- orders_items (fact table) — one row per order line, holding quantity and Total_revenue, related to products to bring in category, cost and stock information.
- orders (dimension) — one row per order, holding order_status, order_year and customer_id, related one-to-many to orders_items.
- products (dimension) — one row per product, holding product_name, category_name and cogs, related one-to-many to orders_items and to inventory.
- payments (dimension/fact) — one row per payment attempt, holding payment_status, related to orders on order_id to compute success and refund KPIs.
- shipping (dimension/fact) — one row per shipment, holding ship_date and return_date, related to orders on order_id to compute fulfilment, on-time shipping and return-rate KPIs.
- customers (dimension) — one row per customer, holding state/city, related one-to-many to orders to enable state-level revenue and demand analysis.
- inventory (dimension) — one row per product, holding current stock, used independently to flag stock-out risk.
- Relationships were set to single-direction, one-to-many from each dimension table into the relevant fact table, keeping the model performant and avoiding ambiguous filter paths.

DAX Measures and KPIs:

1. Total Orders:

Counts the total number of orders placed, used as the headline order-volume KPI.

DAX Measure:

```
1 Total_orders = COUNT('amazon_db orders'[order_id])
```

2. Total Customers:

Counts the distinct number of customers who have placed an order.

DAX Measure:

```
1 Total_customers = DISTINCTCOUNT('amazon_db customers'[customer_id])
```

3. Total Customers per Year:

Computes the distinct customer count for the year(s) currently in filter context, used to build the year-over-year customer trend.

DAX Measure:

```
1 Total Customers per Year =  
2 CALCULATE(  
3     DISTINCTCOUNT('amazon_db orders'[customer_id]),  
4     VALUES('amazon_db orders'[customer_id])  
5 )  
6
```

4. Payment Count:

Counts the total number of payment records, used as the base for the payment success-rate calculation.

DAX Measure:

```
1 Payment Count = COUNT('amazon_db payments'[payment_id])  
2
```

5. Payment Success Rate:

Divides the number of successful payments by total payments to give the overall payment success percentage.

DAX Measure:

```
1 Payment Success Rate =  
2 DIVIDE(  
3     COUNTROWS(FILTER('amazon_db payments', 'amazon_db payments'[payment_status]= "Payment Succeeded")),  
4     COUNTROWS('amazon_db payments'),  
5     0  
6 )  
7
```

6. Refund Rate by Count (%):

Divides refunded orders by total orders to track what share of orders end up refunded, by year.

DAX Measure:

measure should be stored in.

```
1 Refund Rate by Count (%) =  
2 DIVIDE(  
3     [Refunded Orders Count],  
4     [Total_orders],  
5     0  
6 )
```

7. total_cogs:

Calculates total cost of goods sold by multiplying each order line's quantity by the related product's cost, summed across all lines.

DAX Measure:

```
1 total_cogs = SUMX('amazon_db orders_items', 'amazon_db orders_items'[quantity]*RELATED('amazon_db products'[cogs]))
```

8. Gross Profit:

Subtracts total cost of goods sold from total revenue to give gross profit in currency terms.

DAX Measure:

```
1 Gross Profit = 'amazon_db orders_items'[Total_revenue]-'amazon_db orders_items'[total_cogs]
```

9. Average_order_value:

Divides total revenue by total orders to give the average value of a single order.

DAX Measure:

```
1 Average_order_value = DIVIDE('amazon_db orders_items'[Total_revenue] , 'amazon_db orders'[Total_orders])
```

10. Order Fulfillment Rate (%):

Divides the number of orders with status "Completed" by total orders to measure how reliably orders are fulfilled.

DAX Measure:

```
1 Order Fulfillment Rate (%) =  
2 VAR TotalOrders = COUNTROWS('amazon_db orders')  
3 VAR FulfilledOrders= CALCULATE(COUNTROWS('amazon_db orders'),  
4 FILTER('amazon_db orders', 'amazon_db orders'[order_status] = "Completed"))  
5 )  
6 RETURN  
7 DIVIDE(FulfilledOrders,TotalOrders,0)
```

11. Order Return Rate (%):

Divides orders that have a non-blank return_date by total orders to measure the share of orders that come back as returns.

DAX Measure:

```
1 Order Return Rate (%) =
2 VAR ReturnOrders=CALCULATE(COUNTROWS('amazon_db shipping'),
3 NOT(ISBLANK('amazon_db shipping'[return_date])))
4 VAR TotalOrders=COUNTROWS('amazon_db orders')
5 RETURN DIVIDE(ReturnOrders, TotalOrders,0)
```

12. Stockout Alert:

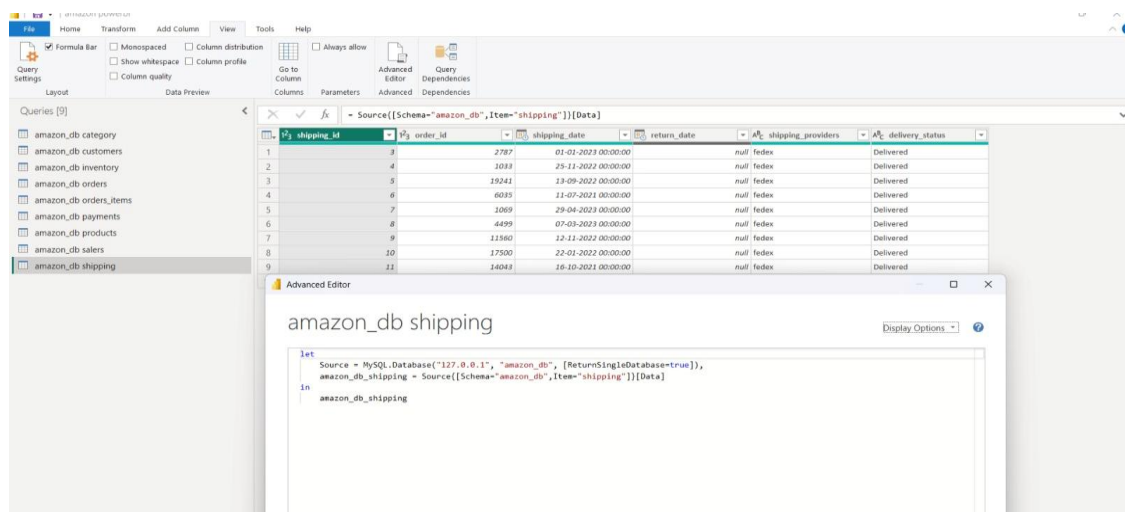
Counts the number of inventory records where stock has fallen to 10 units or below, used to flag products at risk of stocking out.

DAX Measure:

```
1 Stockout Alert =
2 CALCULATE(
3     COUNTROWS('amazon_db inventory'),
4     FILTER(
5         'amazon_db inventory',
6         'amazon_db inventory'[stock]<= 10
7     )
8 )
```

MySQL Server Connection with Power BI Desktop:

- Imported the amazon_db tables (category, customers, inventory, orders, orders_items, payments, products, salers and shipping) from the MySQL database into Power BI using Get Data → MySQL Database.
- Connected via MySQL.Database("127.0.0.1", "amazon_db", [ReturnSingleDatabase=true]) in Power Query, with each table loaded through its own Advanced Editor query (e.g. amazon_db_shipping) referencing the relevant schema and item.
- Enabled the dashboard to work directly off live database data with refreshable connections, rather than a static file extract.

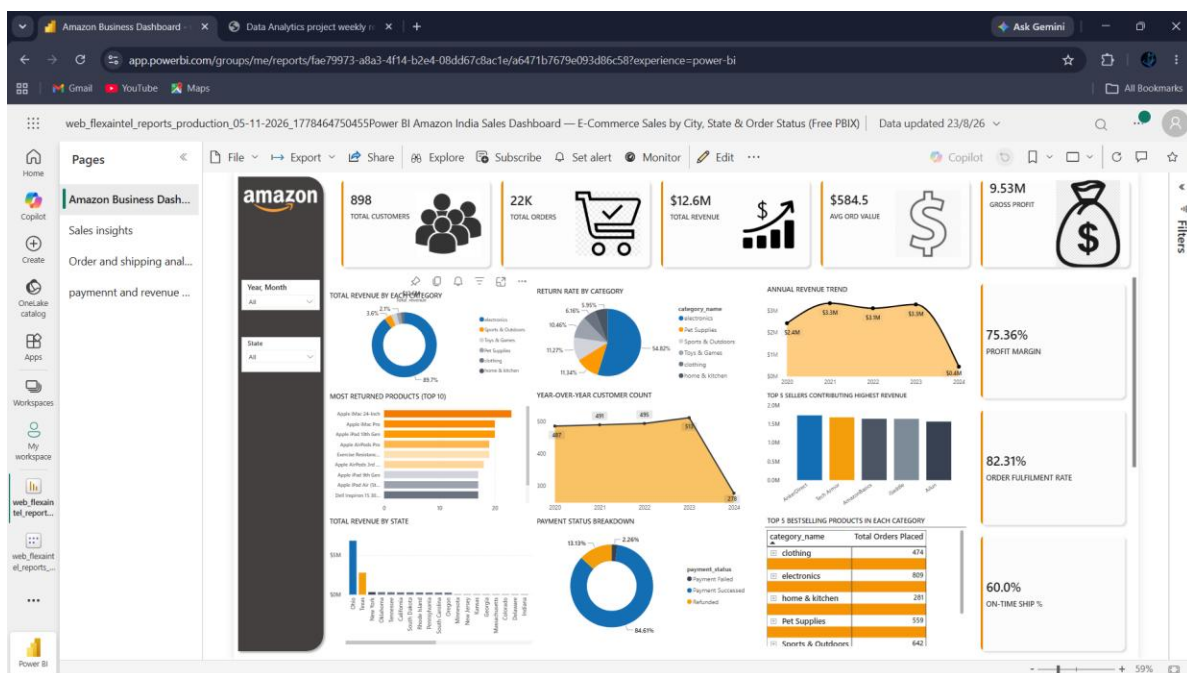


Interactive Amazon Sales Dashboard:

The Power BI report is built across four connected pages, each targeted at a different business audience:

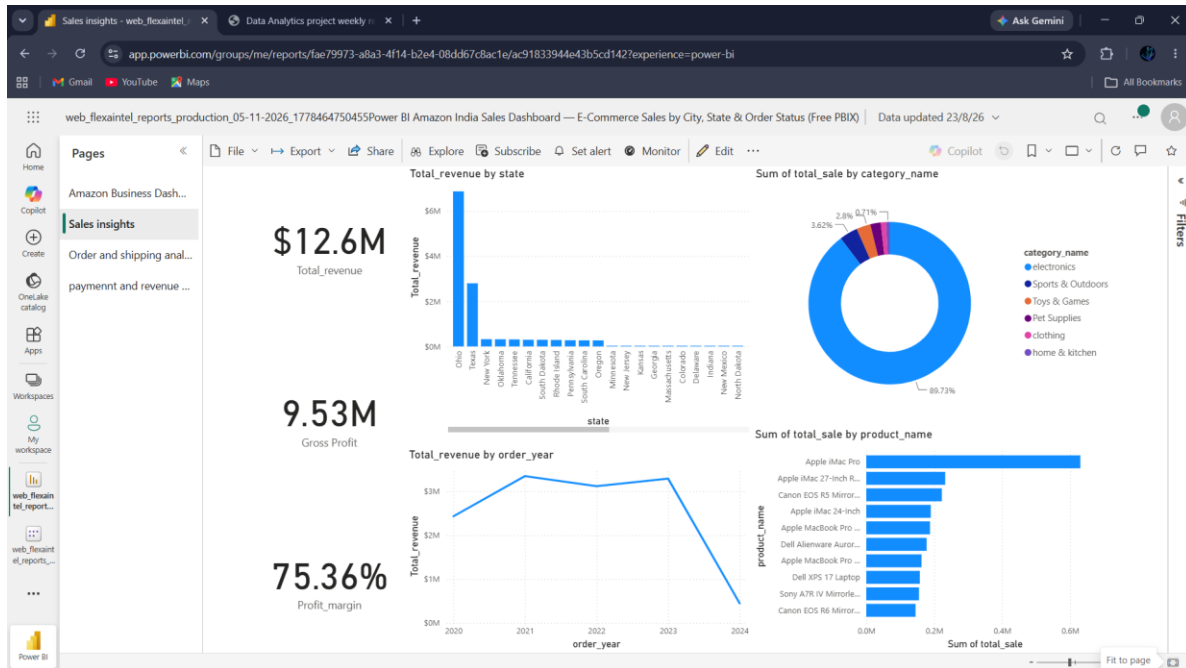
A. Amazon Business Dashboard (Overview)

- Company-wide overview showing 898 total customers, 22K total orders, \$12.6M total revenue, \$584.5 average order value and \$9.53M gross profit.
- Combines return rate by category, most-returned products, year-over-year customer count, annual revenue trend, top 5 sellers by revenue and top 5 bestselling products per category into a single executive view.
- Surfaces headline efficiency KPIs — 75.36% profit margin, 82.31% order fulfilment rate and 60.0% on-time shipping — alongside payment status breakdown and revenue by state.
- Slicers for Year/Month and State let the consultant drill the entire page down to a specific period or region.



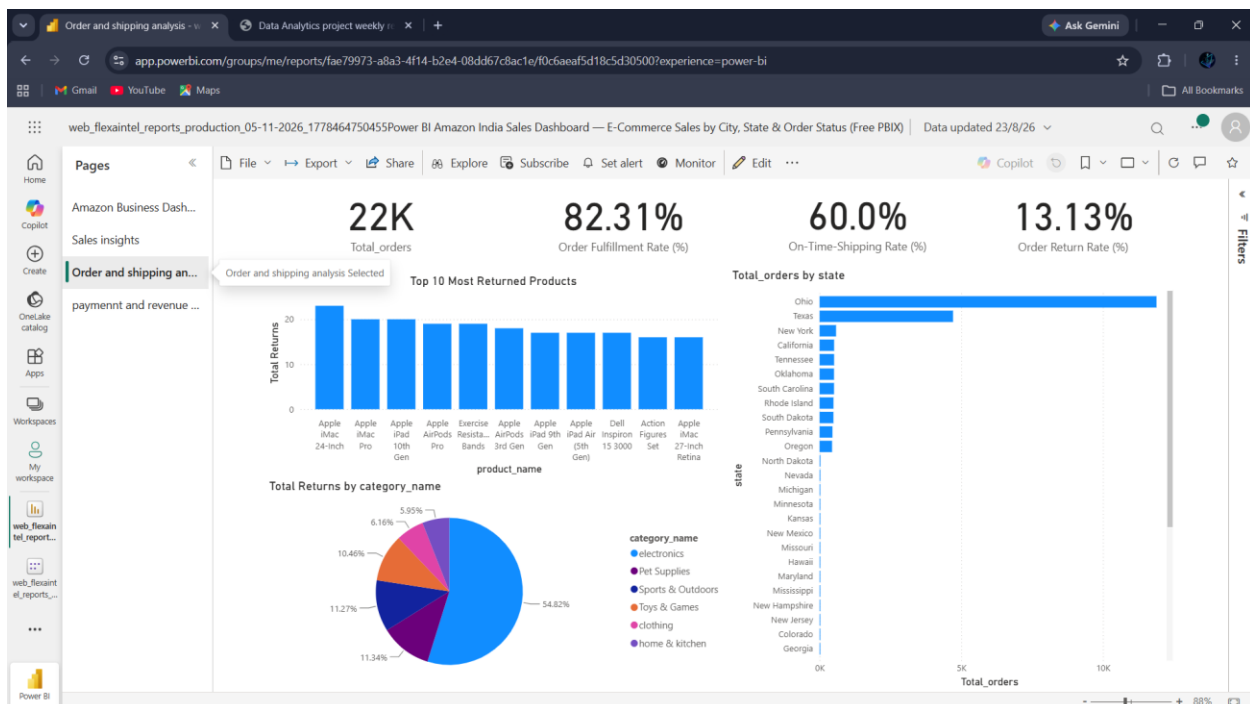
B. Sales Insights

- Focuses on revenue and profitability: total revenue by state, revenue by order year, and gross profit (\$9.53M) and profit margin (75.36%) as standalone KPI cards.
- Shows category-level share of sales via a donut chart (electronics alone accounts for 89.73% of the value shown) and a product-level ranking led by the Apple iMac Pro.
- Revenue by order year shows growth from 2020 through a 2022 peak before declining toward 2024 (a partial year), highlighting where category managers should focus retention efforts.



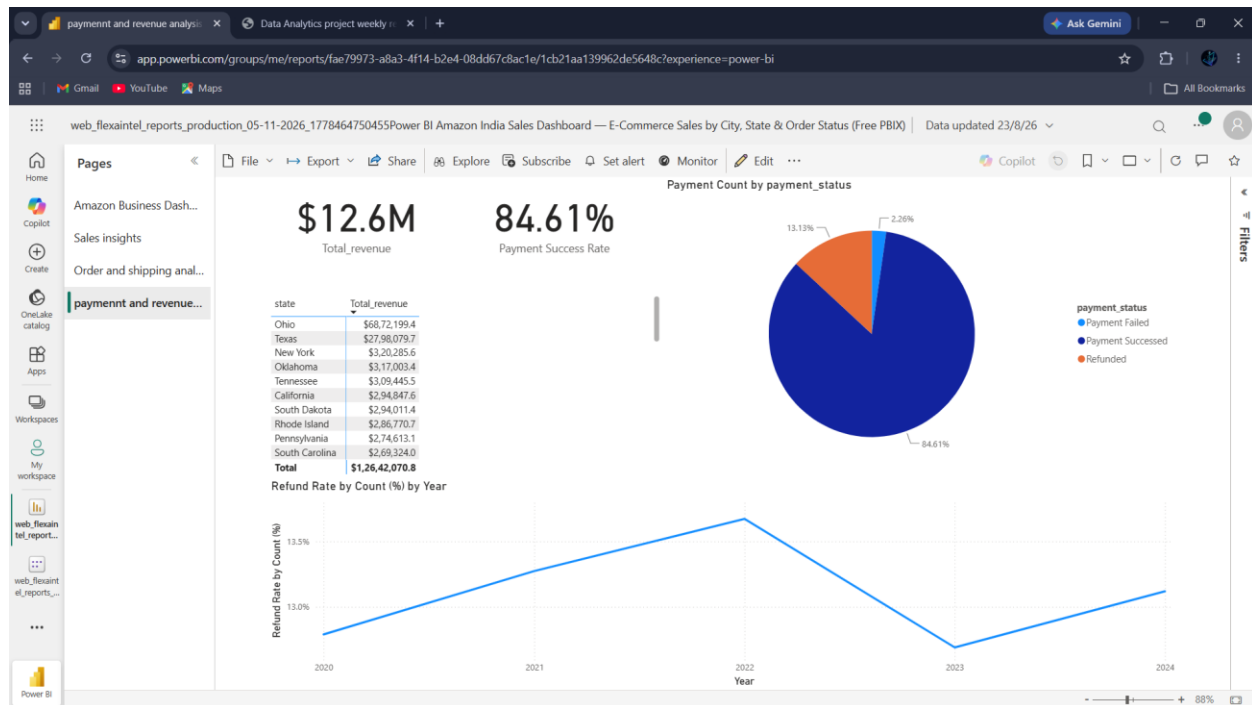
C. Order and Shipping Analysis

- Tracks fulfilment health: 22K total orders, 82.31% order fulfilment rate, 60.0% on-time shipping rate and 13.13% order return rate.
- Top 10 Most Returned Products is led by the Apple iMac 24-Inch, and Total Returns by category shows electronics responsible for 54.82% of all returns.
- Total_orders by state confirms Ohio and Texas as the two largest order-volume states, matching the revenue concentration seen on the Sales Insights page.



D. Payment and Revenue Analysis

- Total revenue of \$12.6M with an 84.61% payment success rate; the payment-status pie chart shows 13.13% refunded and 2.26% failed.
- A state-level revenue table ranks Ohio (\$68,72,199.4) and Texas (\$27,98,079.7) well ahead of all other states.
- Refund Rate by Count (%) by Year shows refunds rising from 2020 to a peak around 2022, dropping sharply in 2023, and rising again toward 2024 — a trend worth investigating with the quality/logistics team.



Business Question Answered:

I. Which states generate the highest revenue, and how concentrated is that revenue?

Answer: Ohio (\$68,72,199.4) and Texas (\$27,98,079.7) together dominate the \$1,26,42,070.8 shown in the top-10 state table, with every other state (New York, Oklahoma, Tennessee, California and others) contributing well under \$35,00,000 each. Revenue is highly concentrated in a small number of states.

II. How healthy is the payment process, and how much revenue is at risk from refunds and failures?

Answer: The overall Payment Success Rate is 84.61%, with 13.13% of payments Refunded and 2.26% Failed. Refunds are a materially larger drag on revenue than outright payment failures.

III. How has the refund rate trended over time?

Answer: Refund Rate by Count (%) by Year rose from about 13.0% in 2020 to a peak above 13.5% in 2022, fell sharply to roughly 13.0% in 2023, and has been climbing again toward 2024. The 2022 peak and the 2023 dip both warrant a closer look at what changed in process or product quality.

IV. How reliably are orders fulfilled and shipped, and how many come back as returns?

Answer: Order Fulfilment Rate stands at 82.31% and On-Time-Shipping Rate at 60.0%, while the Order Return Rate is 13.13% of all orders — meaning roughly 1 in 6 orders is either not completed and shipped on time, and about 1 in 8 comes back as a return.

V. Which products and categories drive the most returns?

Answer: The Apple iMac 24-Inch tops the Most Returned Products list, followed closely by the Apple iMac Pro and several other Apple and Dell electronics items. At category level, electronics alone accounts for 54.82% of total returns — far ahead of Pet Supplies (11.34%), Sports & Outdoors (11.27%) and Toys & Games (10.46%).

VI. How profitable is the business overall?

Answer: Against \$12.6M in total revenue, Gross Profit is \$9.53M, giving a Profit Margin of 75.36%, with an Average Order Value of \$584.5. Profitability is strong once cost of goods sold is accounted for.

VII. How has the customer base grown year over year?

Answer: Year-over-year customer count rose from 487 in 2020 to a peak of 513 in 2023, before dropping to 278 in 2024 — most likely reflecting a partial/incomplete year of data rather than an actual decline in the customer base.

VIII. Which categories and sellers contribute the most order volume and revenue?

Answer: Electronics leads on order count (809 orders placed), ahead of Sports & Outdoors (642), Pet Supplies (559), clothing (474) and home & kitchen (281). On the seller side, AnkerDirect, TechArmor, AmazonBasics, iSaddle and Alian are the top five sellers contributing the highest revenue.

IX. Does revenue concentration by category match the returns picture?

Answer: Yes — electronics contributes about 89.7% of total product sales value on the Sales Insights page, and it is also the category responsible for the majority (54.82%) of returns. The category driving most of the revenue is also the one carrying the most return risk, so it deserves the closest quality and logistics monitoring.

Business Insights and Recommendation:

Key Insights:

- Revenue is heavily concentrated in two states (Ohio and Texas), which together dwarf every other state in the top-10 revenue table — a classic long-tail geography pattern.
- Electronics is both the star performer (~90% of product sales value, top order count) and the biggest return risk (54.82% of all returns), led by Apple iMac and iPad models.
- The gap between Order Fulfilment Rate (82.31%) and On-Time-Shipping Rate (60.0%) shows that shipping speed, not order completion, is the bigger operational bottleneck.
- Refunds (13.13%) are a far larger source of payment-related revenue leakage than outright payment failures (2.26%), and the refund rate has been volatile year to year rather than steadily improving.
- Profitability is strong at the aggregate level (75.36% margin), but this is measured before accounting for shipping, returns-handling and refund costs, which are concentrated in electronics.
- Customer growth appears to have peaked in 2023 (513 customers) before the most recent period, though this needs confirming against whether the latest year of data is complete.

Recommendations:

- Prioritise a root-cause review of electronics returns (starting with the Apple iMac 24-Inch and iMac Pro) — packaging, product description accuracy and quality control are the most likely levers.
- Investigate the On-Time-Shipping gap (60.0% vs 82.31% fulfilment) with the logistics/carrier team, since closing this gap has a direct, low-risk impact on customer satisfaction and repeat purchase.
- Treat Ohio and Texas as core markets for inventory pre-positioning and marketing spend, while testing targeted campaigns to grow the long tail of smaller states.
- Track the Refund Rate by Count (%) trend monthly rather than yearly, so the team can react to spikes (like the 2022 peak) closer to when they happen rather than after a full year.
- Use the Stockout Alert measure operationally — set a review cadence for any product at or below 10 units in stock, particularly within the high-revenue electronics category.
- Re-cut Gross Profit and Profit Margin net of estimated returns and refund cost, to get a more conservative, decision-ready profitability figure for category-level planning.

Conclusion:

This project delivers an end-to-end Business Intelligence solution for the e-commerce retail domain: the raw Amazon order, payment, shipping, product, customer and inventory tables were cleaned and related in Power Query, modelled into a star-schema style structure centred on the orders_items fact table, and enriched with purpose-built DAX measures covering revenue, profitability, order fulfilment, shipping performance, returns, payments and inventory risk. The resulting four-page interactive Power BI dashboard lets a user drill from a company-wide view of 898 customers, 22K orders and \$12.6M revenue down to a single state, category, product or year in a few clicks — answering all the business questions above and surfacing insights that can directly inform logistics planning, category management, payment-risk monitoring and revenue growth strategy.